

Software Testing

What is Software Testing

- *ISTQB / ISEB Foundation Exam Practice*
- *<https://www.guru99.com/software-testing.html>*

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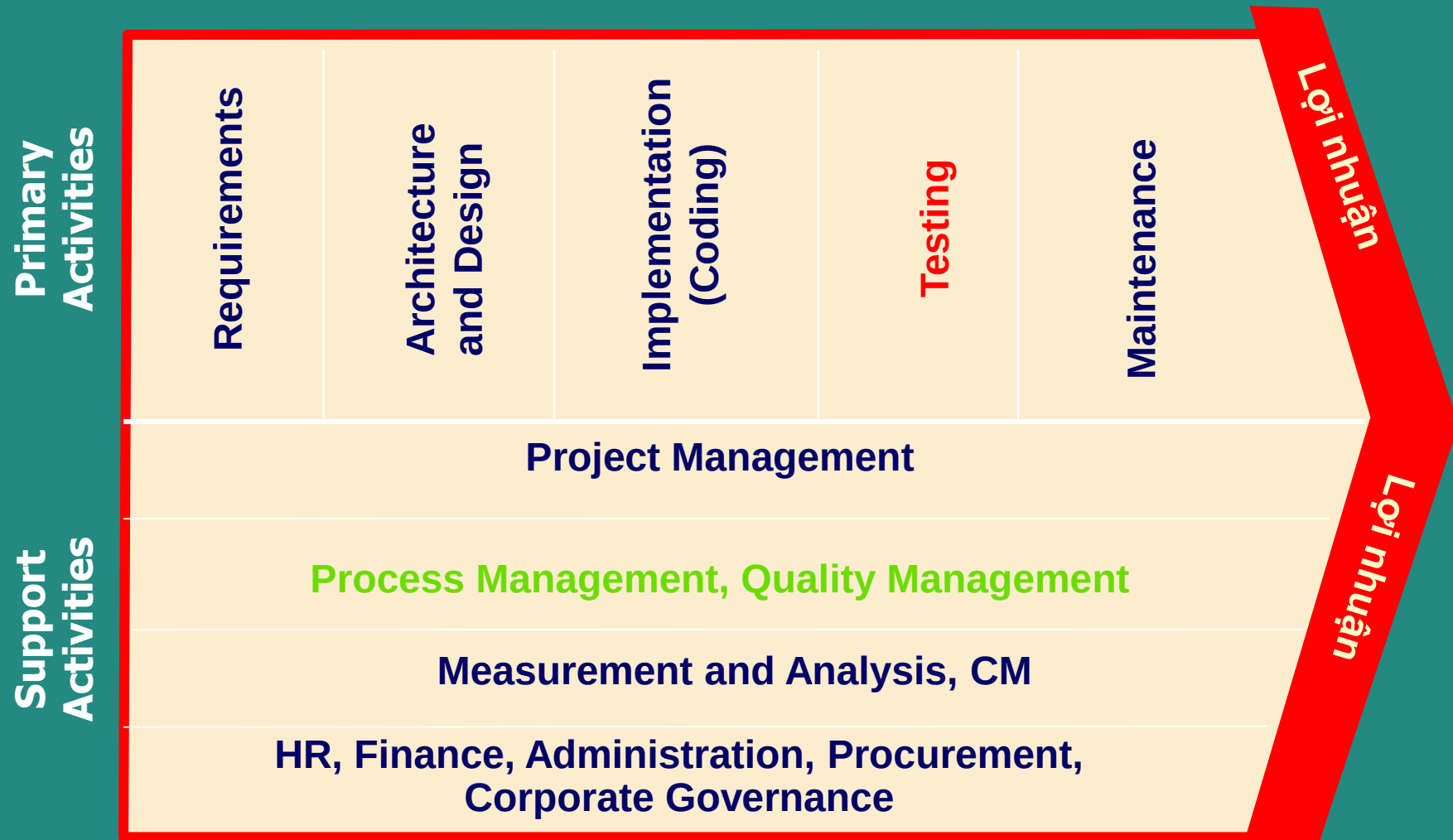
What is software testing

Why testing is necessary

How to become software tester

Certification exams

Software Value Chain



Software Testing

- Software Testing is a method to **check** whether the **actual** software product **matches** expected **requirements**.

Primary Objectives of Testing

1. Ensure that the Application Under Test (AUT) fulfills all requirements of customer
2. Find out the defects or issues occurring in the application before it is encountered by the end-user
3. Evaluate the overall performance of the application
4. Provide a high quality product to the end user

What Does a Software Tester do?

1. understanding requirement documents,
2. creating test cases,
3. **executing test cases,**
4. reporting and re-testing bugs,
5. *attending review meetings ...*

Technical Skills

- Basic knowledge of **Database/ SQL**
- Basic knowledge of **Linux commands**
- Knowledge and hands-on experience of a **Test Management tool**
- Knowledge and hands-on experience of any **Defect Tracking tool**
- Knowledge and hands-on experience of **Automation tool**

Why testers should know programming

- To understand how programs are built
- To get insights of when programs can fail
- To automate some tasks that can help testing
- To understand creative side of developer

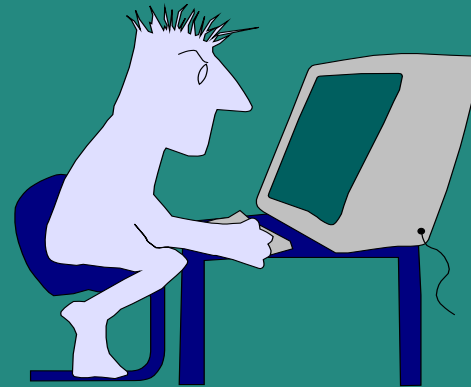
Academic Background

- **ISTQB** Certification: International Software Testing Qualifications Board.
- **CSTE** Certification: Certified Software Test Engineer.

Who does the software testing?



Developer



Tester

Understands the system but, will test "gently" and, is driven by "delivery"



Must learn about the system, but, will attempt to break it and, is driven by "quality"

Definition of some basic terms

- errors, faults (defects) and failures
- testing and debugging
- test case and test suite
- verification and validation
- QC and QA

Errors, Faults (Defects) and Failures

- An error is a *mistake*, misconception, or misunderstanding on the part of a software developer.
- A fault (defect) is introduced into the software as the *result of an error*. It is an anomaly in the software that may cause it to behave incorrectly, and not according to its specification.
- A failure is the *inability* of a software system or component to perform its required functions within specified performance requirements.

Error - Fault - Failure

- **Error**: a human action that produces an incorrect result
- **Fault**: a manifestation of an error in software
 - also known as a defect or bug
 - if executed, a fault may cause a failure
- **Failure**: deviation of the software from its expected delivery or service
 - (found defect)

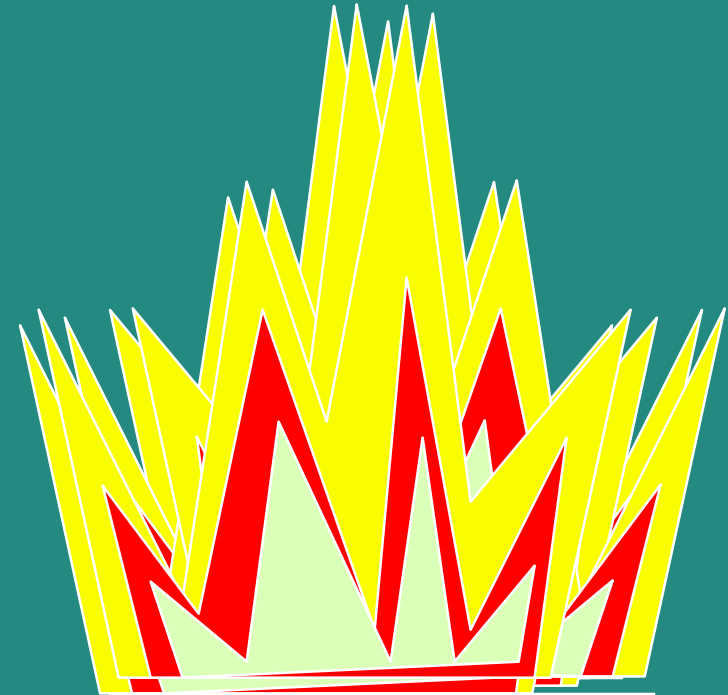
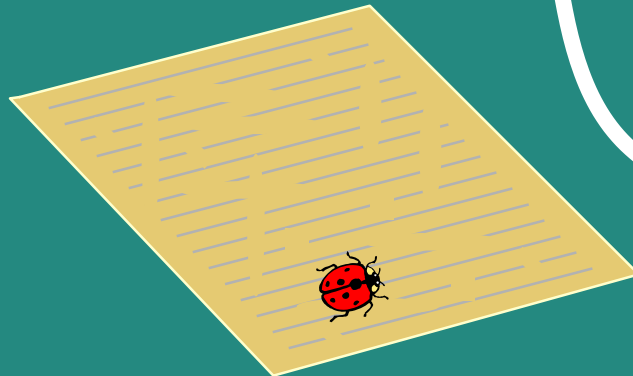
Failure is an event; fault is a state of the software, caused by an error

Error - Fault - Failure

A person makes
an **error** ...



... that creates a
fault in the
software ...
also known as a
defect or **bug**



... that can cause
a **failure**
in operation

Test Case and Test Suite

- A test case is a specification of the inputs, execution conditions, testing procedure, and expected results to verify compliance with a specific requirement.
- A group of related tests that are associated with a database, and are usually *run together*, is sometimes referred to as a test suite.

Verification and Validation

- **Verification** is the process of evaluating a software system or component to determine whether the products of a given development phase *satisfy the conditions* imposed at the start of that phase.
 - Verification answers for the following question:
”Are we building the product right?”
- **Validation** is the process of evaluating a software system or component during, or at the end of, the development cycle in order to determine whether it *satisfies specified requirements*.
 - Validation answers for the following question:
”Are we building the right product?”

Testing vs. Debugging (1)

- At its core, testing is a process of finding "unknown" faults (defects) whereas debugging is a process of removing "known" faults (defects).
- Testing as a process has following aspects: economic, technical, and managerial
- Testing itself is related to two other processes called verification and validation

Testing vs. Debugging (2)

- **Debugging, or fault localization is the process of**
 - 1. locating the fault or defect**
 - 2. repairing the code**
 - 3. retesting the code**
- **The debugging process begins after testing has been carried out and the tester has noted that the software is not behaving as specified.**

QA vs. QC

Quality Assurance Vs Quality Control

QUALITY ASSURANCE

Focus on the prevention of defects

Proactive process

Process-based approach

Manages Quality

QUALITY CONTROL

Focus on the identification of defects

Reactive process

Product-based approach

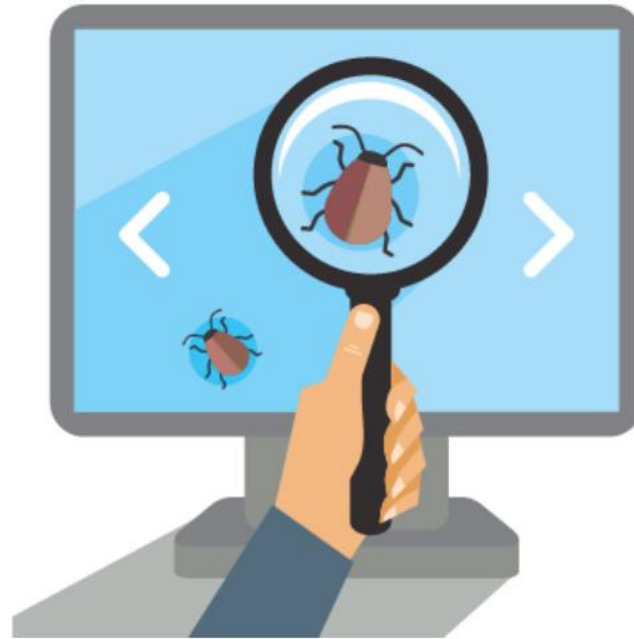
Verify the Quality

Software Testing Myths



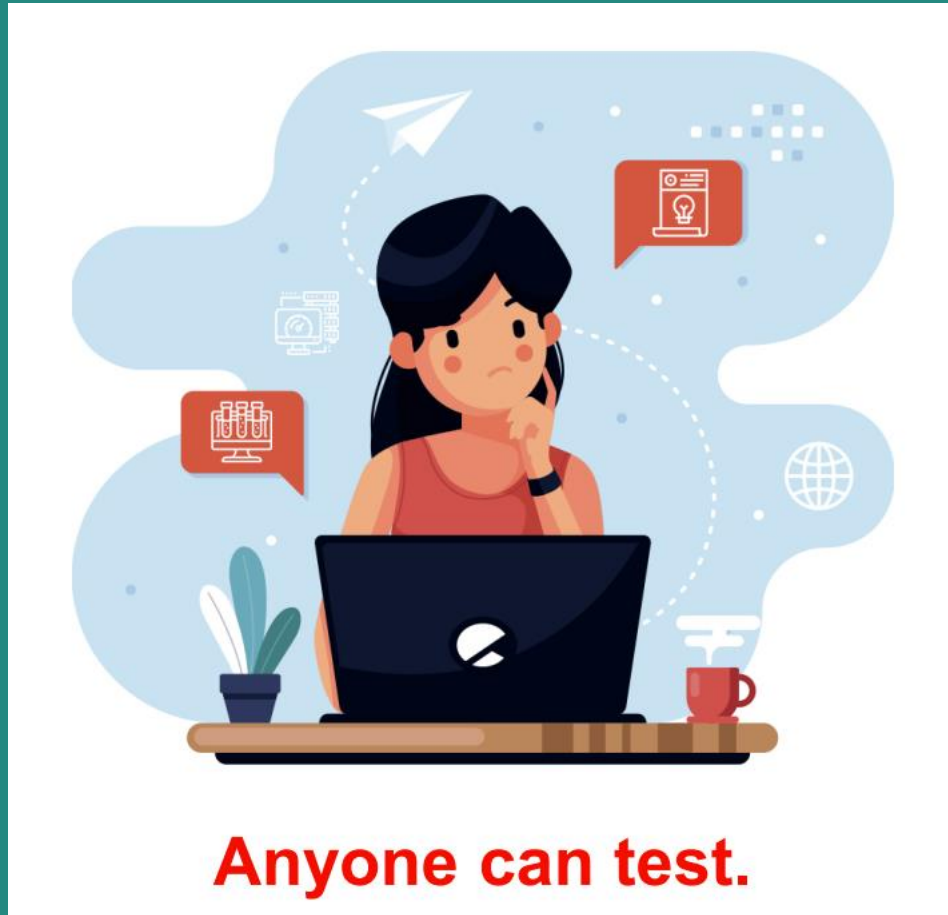
**The objective of Testing is to ensure a
100% defect-free product.**

Software Testing Myths



Testing is easy.

Software Testing Myths

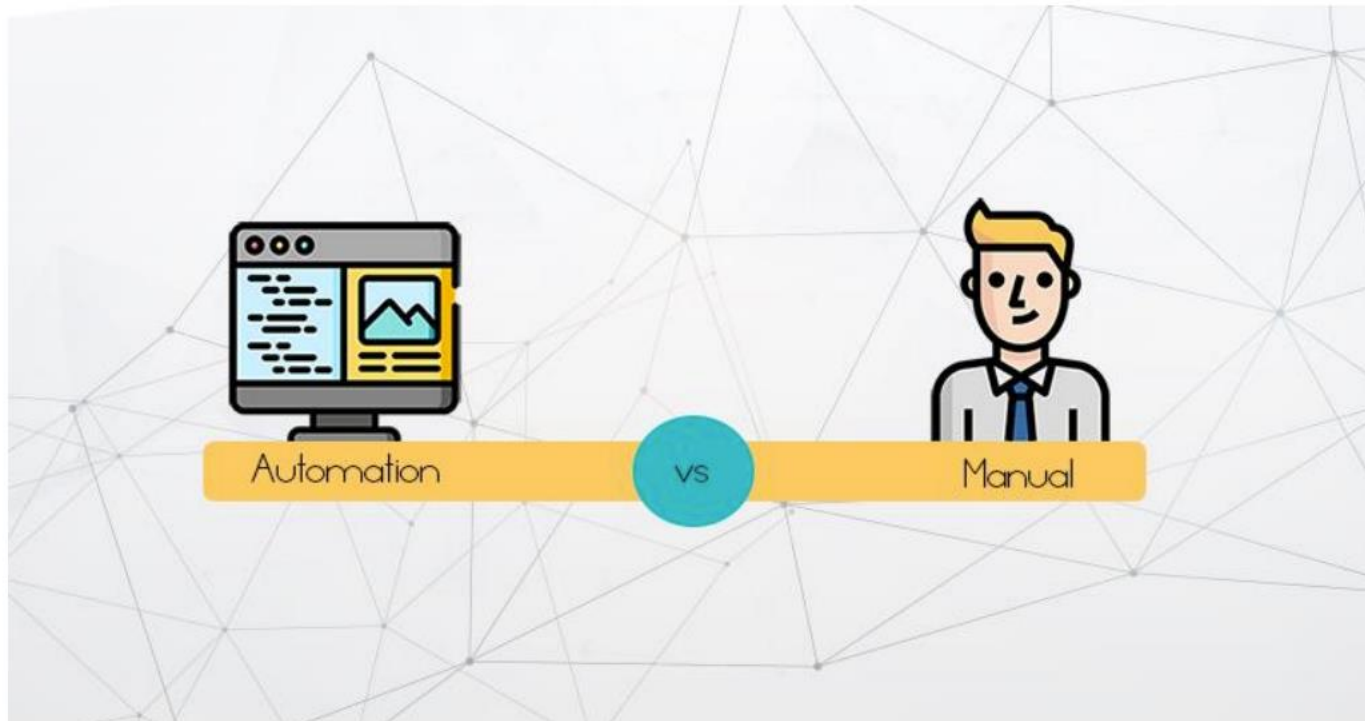


Software Testing Myths



There is no creativity in testing.

Software Testing Myths



Automated testing eliminates the need for manual testing.

Software Testing Myths



When a defect slips, it is the fault of the Testers.

Software Testing Myths



Software Testing does not offer opportunities for career growth.

